INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

LOADING STANDARDS, LIQUID TRANSFER, BULK:

LIQUID FUELS, STATIONARY, 3-INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers stationary loading standards for transfer of bulk liquid petroleum fuels.

1.2 Classification. The loading standards shall be of the following types, as specified (see 6.2 and 6.8):

Type I	-	Loading standard, liquid transfer, bulk: stationary, 3-inch, rigid assembly.
Type II	-	Loading standard, liquid transfer, bulk: stationary, 3-inch, sectional assembly.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3990

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1898	-	Cushioning Material, Packaging (Cellulosic, Water Absorbent).
W-C-440	-	Clip, Electrical.
W-R-550	-	Rod, Ground (With Attachments).
WW-T-700	-	Tube, Aluminum Alloy, Drawn, Seamless, 5052.
WW-T-700/6	-	Tube, Aluminum Alloy, Drawn, Seamless, 6061.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-T-97	-	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-F-3541	-	Fittings, Lubrication.
MIL-G-3859	-	Grease Gun, Hand Operated, Lever, Push and Screw Type.
MIL-A-7021	-	Asbestos Rubber Compressed Sheet, for Fuel, Lubricant, Coolant, Water, and Temperature Resistant Gaskets.
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-G-10924	-	Grease, Automotive and Artillery.
MIL-V-12003	-	Valves, Plug: Cast-Iron or Steel, Manually Operated.
MIL-G-23827	-	Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354, Metric.
MIL-R-24049	-	Rope, Polypropylene.
MIL-C-27487	-	Coupling Halves, Quick-Disconnect, Cam-Locking Type.

STANDARDS

FEDERAL

FED-STD-H28	-	Screw-Thread Standards for Federal Services.

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MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MS27029	-	Coupling Half, Quick Disconnect, Cam-Locking Type, Plug, Dust, Type X.
MS27030	-	Gasket-Coupling Half, Quick-Disconnecting, Cam-Locking Type.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI B16.5 - Steel Pipe Flanges and Flanged Fittings.
ANSI B16.9 - Wrought Steel Buttwelding Fittings (ISO R 285).

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 193	-	Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service.
ASTM A 312	-	Seamless and Welded Austenitic Stainless Steel Pipe.
ASTM A 827	-	Plates, Carbon Steel, for Forging and Similar Applications.
ASTM B 210	-	Aluminum-Alloy Drawn Seamless Tubes.
ASTM B 221	-	Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Shapes and Tubes.
ASTM D 3953	-	Strapping, Flat Steel and Seals.
ASTM D 4675	-	Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

AWS D1.1 - Structural Welding Code-Steel.
AWS D1.2 - Structural Welding Code-Aluminum.
AWS D10.7 - Aluminum Pipe, Gas Shielded-Arc Welding.

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC. AGENT.

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Traffic Order Section, 2200 Mill Rd, Alexandria, VA 22314.)

TRUCK TRAILER MANUFACTURERS ASSOCIATION (TTMA)

TTMA RP28 - Companion Flange.

(Application for copies should be addressed to the Truck Trailer Manufacturers Association, 1020 Princess Street, Alexandria, VA 22314

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, Rm 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The loading standards shall be as shown on figures 1 and 2 and as specified herein. Each loading standard shall dispense fuel at a working pressure of 50 pounds per square inch gage (psig).

3.2 Pressure requirement. The portion of the loading standards between the fuel inlet and the loading valve shall withstand a hydrostatic proof pressure of 150 psig, for not less than 2 minutes, without leakage or permanent deformation or other defects that could harmfully affect the performance and serviceability of the loading standards.

3.3 Human factors engineering. The loading standard shall comply with the design criteria of MIL-STD-1472. Special design emphasis shall be given, but not limited to, MIL-STD-1472, general requirements (4), controls (5.4), labeling (5.5), design for maintainer (5.9), and hazards and safety (5.13), as applicable. The loading standards shall be operable by 5th percentile female through 95th percentile male soldiers, dressed in arctic or NBC protective clothing.

3.4 First article. When specified (see 6.2), the contractor shall furnish a sample unit for first article inspection and approval (see 4.3, and 6.3).

3.5 Materials. Materials shall be as specified herein and as shown on figure 1 and figure 2. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 3.6.6). Materials shall be compatible with all hydrocarbon base (petroleum and synthetic) and alcohol base fuels as well as mixtures of the two. Fuel-wetted parts shall not be fabricated of copper, brass, zinc, bronze or cadmium plating.

3.5.1 Aluminum alloy. Aluminum alloy shall conform to ASTM B 210, Alloy 6061-T6.

3.5.2 Stainless steel. Stainless steel shall conform to ASTM A 312.

3.5.3 Material deterioration prevention and control. The loading standards shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the loading standards may be exposed.

3.5.4 Dissimilar metals. Dissimilar metals, shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

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3.5.5 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.5.6 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the loading standard may be newly fabricated from recovered materials to the maximum extent practicable, provided the loading standard produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the loading standard.

3.6 Buttwelding fittings. Buttwelding fittings shall conform to ANSI B16.9.

3.7 Flanges. Flanges shall be class 150 and shall conform to the requirements specified in ANSI B16.5 and TTMA-RP28; all flanges shall be furnished with the necessary gaskets, bolts, nuts, and washers.

3.8 Gaskets. A gasket shall be furnished for each pair of companion flanges and shall be fabricated of material conforming to MIL-A-7021, class 1, 1/16-inch thick.

3.9 Components.

3.9.1 Loading arm assembly. The loading arm assembly shall be a 3-inch diameter commercial type, equipped with O-ring seals and having the following operational characteristics:

a. Swing freely through 360 degrees in a horizontal plane.
b. Swing freely through a vertical arc from not less than 30 degrees below to 70 degrees above the horizontal plane.
c. When manually lowered to the loading position, the loading arm shall automatically remain in that position during loading. When released from loading position, the loading arm shall be returned to its original position by adjustable counterweights.
d. When swung through its vertical arc, the drop tube shall hang in a vertical position by its own weight.

3.9.1.1 Loading valve. The loading valve shall be a lever operated, self-closing type, of such construction as to assure a smooth shockproof closure and shall be equipped with a vacuum breaker. The valve shall function without failure and shall not leak or show any indication of structural failure when subjected to a hydrostatic proof pressure of 150 psig, for not less than 2 minutes. The valve control shall be labeled to indicate function and shall be provided with a double-ended arrow showing direction of control

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operation, labeled at each end to indicate the functional result (i.e. open, close, etc.).

3.9.1.2 Remote-control handle. A remote-control handle shall be connected to the loading valve lever by means of a metal connecting rod as shown in figure 2; or when specified (see 6.2), a rope shall be attached to the lever and extending to the handle or eye provided on the single swing joint as shown on figure 1. The rope shall be polypropylene, conforming to MIL-R-24049, type I, 3/8 inch diameter, 12 feet long. The valve control shall be labeled to indicate function and shall be provided with a double-ended arrow showing direction of control operation, labeled at each end to indicate the functional result (i.e. open, close, etc.).

3.9.2 Sliding tube. The sliding tube shall be an assembly of two telescoping tubes. The outer tube shall be seamless, drawn, round aluminum tubing conforming to WW-T-700 and WW-T-700/6, type I, temper T6 or extruded, round aluminum tubing conforming to ASTM B 221, 6063-T832. The inner tube shall be aluminum or 300 series stainless steel with an inside diameter of not less than 3 inches. The tube shall be fully guided and shall slide easily from approximately the 5-foot telescoped position to the 9-foot fully extended position. One end of the assembly shall be externally threaded. The other end shall have TTMA RP28 flange welded on. The outer tube shall have a stuffing box with a packing compatible with petroleum fuels. The stuffing box shall be equipped with a packing gland and a lugged and threaded take-up ring for adjustment. The sliding tube assembly including the drop tube, single swing joint, and the outside tube stuffing box, shall not leak when subjected to a 25 psig hydrostatic proof pressure for 5 minutes.

3.9.3 Swivel joint, double. The double swivel joint shall be equipped with double rows of ball bearings for each rotative direction. The bearings shall be sealed to prevent leakage and entry of fuels into the bearing cavities, and shall be provided with lubrication fittings specified in 3.10.1. One end of the joint shall have a TTMA RP28 flange welded on, the other-inlet-end shall have class 150 flange. The double swivel joint shall provide a 360-degree rotation in both the horizontal and the vertical planes.

3.9.4 Swivel joint, single. The single swivel joint shall consist of two aluminum 90-degree elbows, grooved on the inside, to receive O-ring seals, and on the outside to receive a split retainer collar to prevent disjointing. Elbows and O-ring seals shall ride on a sleeve, acting as a trunnion. A handle shall be cast integrally with the elbow which is to receive the drop tube. The inlet and outlet shall be internally threaded. The single swivel joint shall provide a 360-degree rotation in one plane only.

3.9.5 Valves, plug. The plug valves shall conform to MIL-V-12003, type I, class 150, 4-inch size. Pattern shall be venturi or regular opening. Valve ends, inlet and outlet shall be flanged, with class 150 flange.

3.9.6 Grounding reel. The grounding reel shall be a commercial, enclosed type, and shall have the following characteristics and component parts:

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a. Steel cable 3/32-inch, zinc coated, approximately 40 feet in length and a minimum tensile strength rating of 1,000 pounds. The cable shall terminate in an electric power clip with lug conforming to W-C-440, type PCL, nominal current rating, 100 amperes.
b. Automatic rewind for cable.
c. Provisions for vertical mounting on the support bracket.
d. Ratchet device to latch the cable at intervals.
e. Instruction plate specified in 3.12.1.

3.9.7 Rod, ground. The ground rod shall be a driven sectional rod conforming to W-R-550, type III, class B.

3.9.8 Ground connections. A double-terminal ground clamp shall be placed around the riser pipe, approximately as shown on figures 1 and 2. Ground wire for terminals shall be 0.24 inch nominal diameter, No. 4 AWC, stranded copper wire. One 10-foot length and one 5-foot length of ground wire shall be furnished. The grounding connection shall be labeled "GROUND".

3.9.9 Base. The base and gusset plates shall be made from steel plate conforming to ASTM A 827, composition grade UNS G10200, dimensions as shown on figures 1 and 2. The plates shall be welded in place at the end of the standpipe.

3.9.10 Coupling half, inlet connection. The coupling half shall be flanged female cam-locking type quick-disconnect, size 4 inches, equal to OPW 633-LBS coupler, and shall be equipped with a gasket conforming to MS27030-9 and dustplug with chain attachment conforming to MS27029-17. "OPW" refers to the OPW Division of the Dover Corporation.

3.10 Lubrication. All surfaces requiring lubrication shall be provided with means for lubrication.

3.10.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I or III, as applicable. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859 without removal or adjustment of accessories or parts. Fittings shall incorporate provisions for release of excess lubricant to the atmosphere.

3.10.2 Lubricant. Grease lubrication, including lubrication of sealed bearings, shall be with standard military thermally stable greases conforming to MIL-G-10924 or MIL-G-23827, as applicable.

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3.11 Safety requirements.

3.11.1 Static electricity protection. With the loading standard fully erected and ready for operation, and the cable from the grounding reel extended to its full length, there shall be not more than 3 ohms resistance between the tip of the drop tube and the grounding cable attachment clip.

3.11.2 Electric conductivity. Where electric conductivity for static grounding is required, protective coatings and any materials with insulating properties shall be removed from between contacting parts. The flange type coupling, swing joint, and loading arm assembly shall be provided with graphited packing and gaskets, sliding brushes, or other positive means of insuring the electrical conductivity required.

3.11.2.1 Support bracket. The area of the support bracket which comes in contact with the grounding reel shall not be painted or anodized to insure a positive ground.

3.12 Identification marking. The loading standards shall be identified in accordance with MIL-STD-130. The marking shall be applied to the loading standards on plates conforming to MIL-P-514, type III, composition C, of type I, class 1, grade B material, thickness 0.008 to 0.012 inch; black background with aluminum lettering. Plates shall be attached by screws, bolts, or rivets in a location where the plates will be visible and legible.

3.12.1 Instruction plates. The loading standards shall be equipped with instruction plates suitably located, describing any special or important procedure to be followed in operating and servicing the loading standards. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, class 1, material thickness 0.008 to 0.012 inch; black background with aluminum lettering. Plates shall be attached by screws, bolts, or rivets in a location where the plates will be visible and legible. One plate shall be located at the grounding reel in a conspicuous location. This plate shall give the following warning: "Copper grounding clip must be attached to the truck prior to starting loading operations".

3.13 Treatment and painting. The portions of the loading standards normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G. Unless otherwise specified (see 6.2), the color shall be 34088.

3.13.1 Anodic coating. The loading standards and component parts, fabricated from aluminum alloy, when not painted, shall be anodized in accordance with MIL-A-8625, type II, class 2 to a minimum thickness of 0.0004 inch for coatings and 0.0007 inch for wrought. Color shall be as specified (see 6.2).

3.14 Interchangeability. All identical components of each type loading standard shall be interchangeable.

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3.15 Threaded connections. All threaded connections shall conform to FED-STD-H28 and shall be assembled by the contractor. The threaded connections except drop tube shall withstand a hydrostatic pressure of 150 psi for 2 minutes without any evidence of leakage.

3.16 Types of loading standards.

3.16.1 Type I, rigid. The type I, rigid loading standard shall be as shown on figure 1 and as specified herein. Eight hook anchor bolts 5/8 inch diameter by 12 inches long with nuts and washers conforming to ASTM A 193 shall be furnished for anchoring of the base plate to a concrete footing. The concrete shall not be furnished.

3.16.1.1 Coupling half, can-fill. The coupling half shall conform to MIL-C-27487, type III, style B, class 1, size 2 inches, and shall be equipped with a gasket and dust cap with chain attachment conforming to MIL-C-27487, type IX, class 1, size 2-inch. A nipple, threaded to fit the 2-inch male coupling half, shall be welded in place to the 4-inch riser pipe in location as shown on figure 1, at 5 feet above ground level. A hook to hold the can-fill nozzle shall be welded to the riser pipe in the approximate location shown on figure 1.

3.16.1.2 Drop tube. The drop tube shall be aluminum tubing conforming to WW-T-700 and WW-T-700/6, type I, temper T6 or ASTM B 221, 6063-T832. The tube shall be not less than 3 inches inside diameter by 5 feet long. One end of the tube shall be externally threaded for connecting to the single swivel joint. The other end shall be beveled as shown on figure 1; or when specified (see 6.2), shall be fitted with a T-deflector similar to the one specified for the type II loading standard.

3.16.2 Type II, sectional. The type II, sectional loading standard shall be as shown on figure 2, and as specified herein. The concrete shall not be furnished.

3.16.2.1 Riser pipe. The riser pipe shall be in four sections as shown in figure 2, three 6-foot sections and one 5-foot section. When specified (see 6.2), the third (upper) 6-foot section shall not be furnished. This section of the riser pipe is not required when the loading standard is to be used for loading of tank trucks only.

3.16.2.2 Drop tubes. The drop tubes shall be aluminum tubing conforming to WW-T-700 and WW-T-700/6, type I, temper T6 or ASTM B 221, 6063-T832. Inside diameter shall be not less than 3 inches. Two lengths of tube shall be furnished. One tube shall be 5 feet and the other 11 feet in length. Each drop tube shall be equipped on one end with the female half coupling equal to OPW 128-R ring and OPW 132-T tail. The male half coupling shall be equal to OPW 132 and shall be connected to the single swivel joint. The opposite end of each drop tube shall be equipped with an anti-splash deflector equal to OPW 464 deflector. "OPW" refers to the OPW Division of the Dover Corporation. When specified (see 6.2), the 11-foot drop tube shall not be furnished (see 6.1.2).

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3.17 Workmanship. The loading standards shall be free from defects such as misaligned components, incomplete welds, cracks, burns, leaks, and other defects that could impair the operation and serviceability of the loading standards. All parts shall be clean and free from dirt, sand, grease, oil, and metal chips. All external surfaces shall be free of burrs, sharp edges and corners, except where sharp edges or corners are required and not detrimental to safety.

3.17.1 Welders and welding operators. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either AWS D1.1, D1.2, or the ASME code for the materials jointed and the type of welding operation to be performed and that such qualification is effective as defined. Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator (see 6.6).

3.17.2 Welding (aluminum). Welding performed in the fabrication of the loading standards shall be in accordance with AWS D10.7. The surface of parts to be welded shall be free from grit, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

3.17.3 Castings and forgings. All parts, components, and assemblies of the loading standards which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies

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submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance inspection does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and figures, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. When a first article sample is required, the first article sample shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. When a first article sample is required, the first article sample shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Unit of product. For the purpose of inspection, a unit of product shall be all individual parts and components required for assembling one complete loading standard.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when 0 defects are found and rejected when 1 or more defects are found.

4.4.2.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105, level S-2, for both the sliding tube assembly and the complete loading standard. Failure of any test shall be cause for rejection.

4.4.3 Examination. Samples selected in accordance with 4.4.2.1 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4.4 Tests.

4.4.4.1 Sliding tube assembly. Samples selected in accordance with 4.4.2.2 shall be tested as specified in 4.5.2.3.2. Failure of test shall be cause for rejection.

4.4.4.2 Loading standard (complete). Samples selected in accordance with 4.4.2.2 shall be tested as specified in 4.5.2.1, 4.5.2.2, and 4.5.2.3.1. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The loading standards shall be examined for the following defects:

Major

101. Dimensions not as specified (see 3.1).
102. Human factors not as specified (see 3.3).
103. Materials not as specified (see 3.5).
104. Aluminum alloy not as specified (see 3.5.1).
105. Stainless steel not as specified (see 3.5.2).
106. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment (see 3.5.3).
107. Dissimilar metals not as specified (see 3.5.4).
108. Contractor does not have documentation available for identification of material, material finishes or treatments (see 3.5.5).
109. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the loading standard (see 3.5.6).
110. Buttwelding fittings not as specified (see 3.6).
111. Flanges not as specified (see 3.7).
112. Gaskets not as specified (see 3.8).
113. Loading arm assembly not as specified (see 3.9.1).
114. Sliding tube assembly not as specified (see 3.9.2).
115. Swivel joints, double and single not as specified (see 3.9.3 and 3.9.4).
116. Valves, plug not as specified (see 3.9.5).
117. Grounding reel and component parts not as specified (see 3.9.6).
118. Ground rod not as specified (see 3.9.7).
119. Ground connections not as specified (see 3.9.8).
120. Base not as specified (see 3.9.9).
121. Coupling half, inlet connection not as specified (see 3.9.10).
122. Lubrication means and lubricants not as specified (see 3.10).
123. Safety requirements not as specified (see 3.11).
124. Identical components not interchangeable (see 3.14).
125. Threaded connections not as specified (see 3.15).
126. Type I or type II standard not as specified (see 3.16).

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127. Workmanship not as specified (see 3.17).
128. Welders certification not as specified (see 3.17.1).
129. Welding not as specified (see 3.17.2).
130. Castings and forgings not as specified (see 3.17.3).

Minor

201. Lubrication fittings missing or not as specified (see 3.10.1).
202. Identification marking missing, incomplete, or illegible (see 3.12).
203. Instruction plates missing, incomplete, or illegible (see 3.12.1).
204. Treatment and painting not as specified (see 3.13).
205. Anodic coating color not as specified (see 3.13.1).

4.5.2 Tests.

4.5.2.1 Loading arm assembly. Manually turn the loading arm 360 degrees in a horizontal plane, then vertically move the arm from not less than 30 degrees below, to not less than 70 degrees above the horizontal plane, the drop tube remaining in vertical position during testing. Manually lower the loading arm assembly into its loading position and leave for 2 minutes, then release the loading arm assembly and check its return to original position. Nonconformance to 3.10.1a, b, c, or d shall constitute failure of this test.

4.5.2.2 Static, electrical. Measure the electrical resistance between the tip of the drop tube and the grounding cable attachment clip. Nonconformance to 3.11.1 shall constitute failure of this test.

4.5.2.3 Hydrostatic tests.

4.5.2.3.1 Loading standard. Subject the portion of the loading standard, from the fuel inlet to the loading valve, to a hydrostatic proof pressure of 150 psi, with the loading valve in the closed position. Retain the pressure for a minimum period of 2 minutes, rotate the loading arm 360 degrees and check the loading valve, riser pipe, double swing joint, and O-ring seals for leaks. Check vacuum breaker for proper operation. Nonconformance to 3.1, 3.2, 3.9.1.1 or 3.15 shall constitute failure of this test.

4.5.2.3.2 Sliding tube assembly. Subject the sliding tube assembly to a hydrostatic proof pressure of 25 psi and hold for a period of 5 minutes. Nonconformance to 3.9.2 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of packaging.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

131.	Disassembly not as specified for level A (see 5.1.1.1).
132.	Unprotected exterior metal surfaces not coated with preservative as specified for level A (see 5.1.1.2).
133.	Components not lubricated as specified for level A (see 5.1.1.3).
134.	Gaskets, anchor bolts, nuts, washers, and technical publication not preserved as specified for level A (see 5.1.1.4, 5.1.1.5, and 5.1.1.6).
135.	Securing and packaging of miscellaneous components not as specified for level A (see 5.1.1.7).
136.	Shipping containers not closed and sealed as specified for level A (see 5.1.1.7).
137.	Contents not blocked and braced to prevent movement and damage for level A or B (see 5.2.1 and 5.2.2).
138.	Strapping not as specified for level A (see 5.2.1).
139.	Packing not in accordance with referenced document for level C (see 5.2.3).
140.	Marking missing, illegible, incorrect, or incomplete for level A, B, or C (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The loading arm assembly, ground rod, ground wire, and reel shall be disassembled from the riser pipe. Type II loading standards shall have the riser pipe disassembled into its various sections.

5.1.1.2 Unprotected surfaces. Unprotected exterior metal surfaces of components of the loading standards requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.3 Lubrication. Components for which lubrication fittings have been provided shall be lubricated to capacity with the lubricant specified in 3.10.2.

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5.1.1.4 Gaskets. Loose gaskets shall be individually preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.5 Anchor bolts, nuts, and washers. The quantity of anchor bolts, nuts, and washers required shall be preserved together in accordance with MIL-P-116, method IC-2, using the contractor's carton or a close-fitting commercial type paperboard or fiberboard box.

5.1.1.6 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.7 Miscellaneous components. The rope or connecting rod shall be secured to the tube of the loading arm assembly. The drop tube shall be folded against and secured to the loading arm. Securing shall be with soft annealed wire or with tape conforming to PPP-T-97, type IV, size 3/4 inch. The valve shall be protected from damage by cushioning with cushioning material conforming to A-A-1898, grade II, class B, style and size optional, secured in place with tape specified herein. The ground wire shall be coiled and the coil secured. The coiled ground wire, reel, grounding rod, technical publications, gaskets, anchor bolts, nuts, and washers and any small loose components removed during disassembly, shall be placed together in a close-fitting box conforming to PPP-B-636, type CF, class weather resistant, variety and grade as applicable to the weight of the contents to be packaged. The contents shall be cushioned to prevent movement or damage. The box shall be closed and sealed as specified for method V in the appendix to the box specification.

5.1.2 Level C. Each complete disassembled loading standard shall be preserved in a manner to afford protection against deterioration and damage during shipment from the contractor to the initial destination. In addition, components for which lubrication fittings have been provided shall be lubricated to capacity with the lubricant specified in 3.10.2.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The components comprising a complete loading standard, preserved as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style optional. The contents shall be blocked and braced within the box in a manner to prevent movement or damage. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675.

5.2.2 Level B. The components comprising a complete loading standard preserved as specified in 5.1, shall be packed as specified in 5.2.1 for level A except the box shall be domestic type or class as applicable. Box closure shall be in accordance with the appendix to the applicable box specification.

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5.2.3 Level C. The components comprising a complete loading standard, preserved as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking. In addition to any special marking specified in the marking contract or purchase order (see 6.2) shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature which is helpful, but is not mandatory.)

6.1 Intended use. The stationary loading standards are intended for use in filling of tank railroad cars, tank trucks, and tank trailers.

6.1.1 Type I. The type I rigid loading standard is intended for loading tank trucks, tank trailers, and smaller vehicles and cars.

6.1.2 Type II. The type II sectional loading standard is intended for loading of tank railroad cars, tank trailers, and tank trucks. When this standard is required only for loading of tank trucks and tank trailers, the third (upper) 6-foot section of the riser pipe and the 11-foot long drop tube is not required. For tank railroad cars, the top 6-foot section of riser pipe is used together with the 11-foot length of drop tube. When the intended use condition is not definitely known in advance of procurement, the 6-foot section of riser pipe as well as the 11-foot length of drop tube should be specified.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type of loading standard required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When sample units will be required for first article inspection (see 3.4).
e. When rope is required in place of connecting rod (see 3.9.1.2).
f. Finish paint if other than as specified (see 3.13).
g. Color of anodic coating (see 3.13.1).
h. When T-deflector is required on drop tube of type I loading standard (see 3.16.1.2).
i. When the third (upper) 6-foot section of riser pipe is not required (see 3.16.2.1).
j. When the 11-foot drop tube is not required (see 3.16.2.2).
k. Level of preservation and level of packing required (see 5.1 and 5.2).
l. Any special marking (see 5.3).

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6.3 First article. When a first article is required, the items should be a preproduction model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first article. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the soliciation.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary, including any special tools, and instructions regarding shipment of loading standard.

6.6 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
3.17.1	DI-MISC-80875	Welding Procedures	
3.17.1	DI-MISC-80876	Welding Procedure Qualification Test Report	

6.7 Subject term (key word) listing.

Bulk liquid fuels
Petroleum fuels
Loading standard
Liquid transfer

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6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Part or identifying number (PIN). The following specification based part identification numbering procedure shall be used to identify the items covered by this document. The purpose of this procedure is to assist Government catalogers in developing correct equipment/material PINs. This procedure does not constitute a requirement of the contractor (see 1.2).

M3058 X
- X: Type (1 = I, 2 = II)
- M3058: Document identifier

Custodians:
Army - ME
Navy - YD

User activity:
Army - CE

Review activity:
Army - AT
DLA - GS

Preparing activity:
Army - ME

Project 3990-0217

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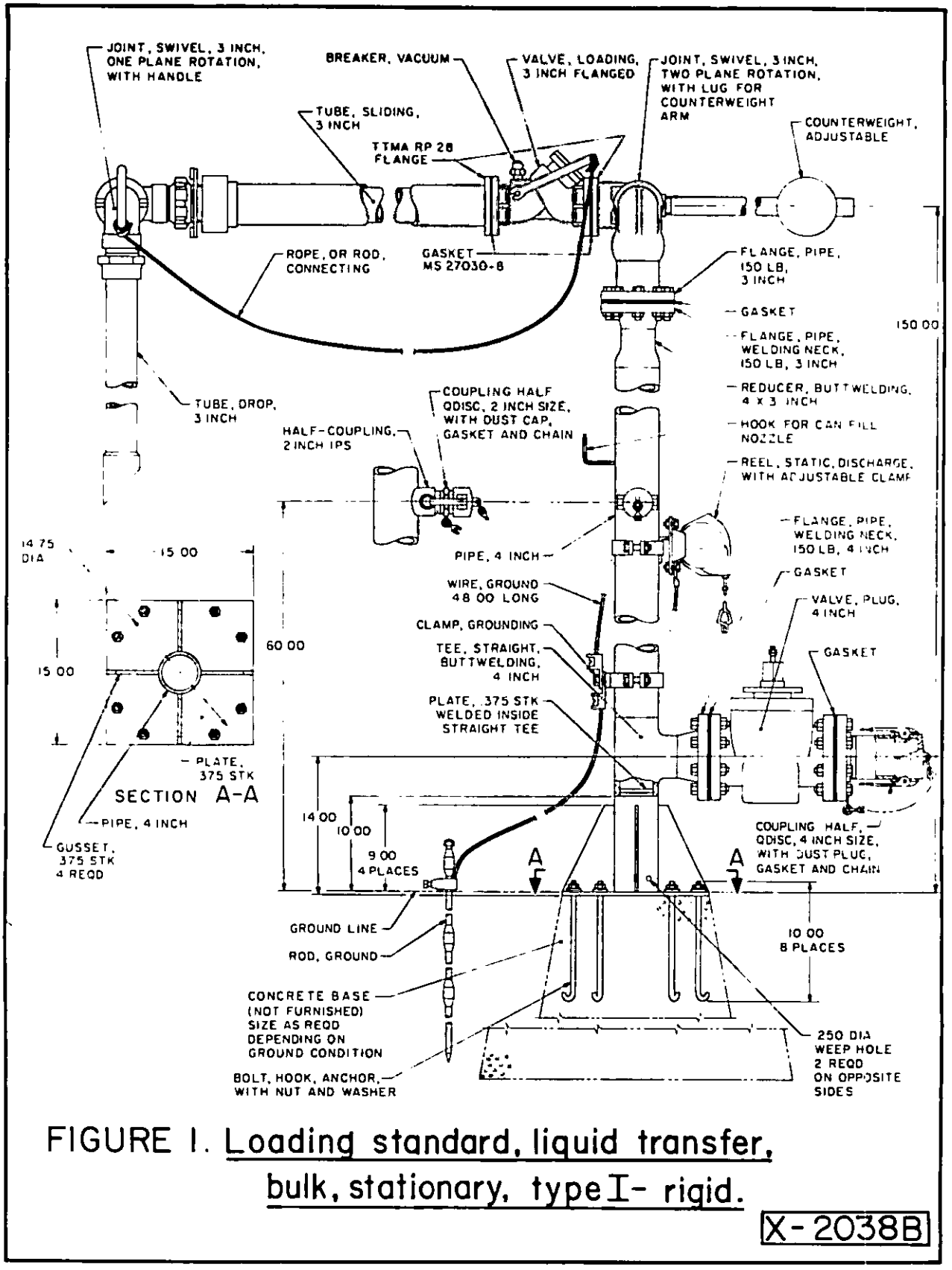


FIGURE 1. Loading standard, liquid transfer, bulk, stationary, type I- rigid.

X-2038B

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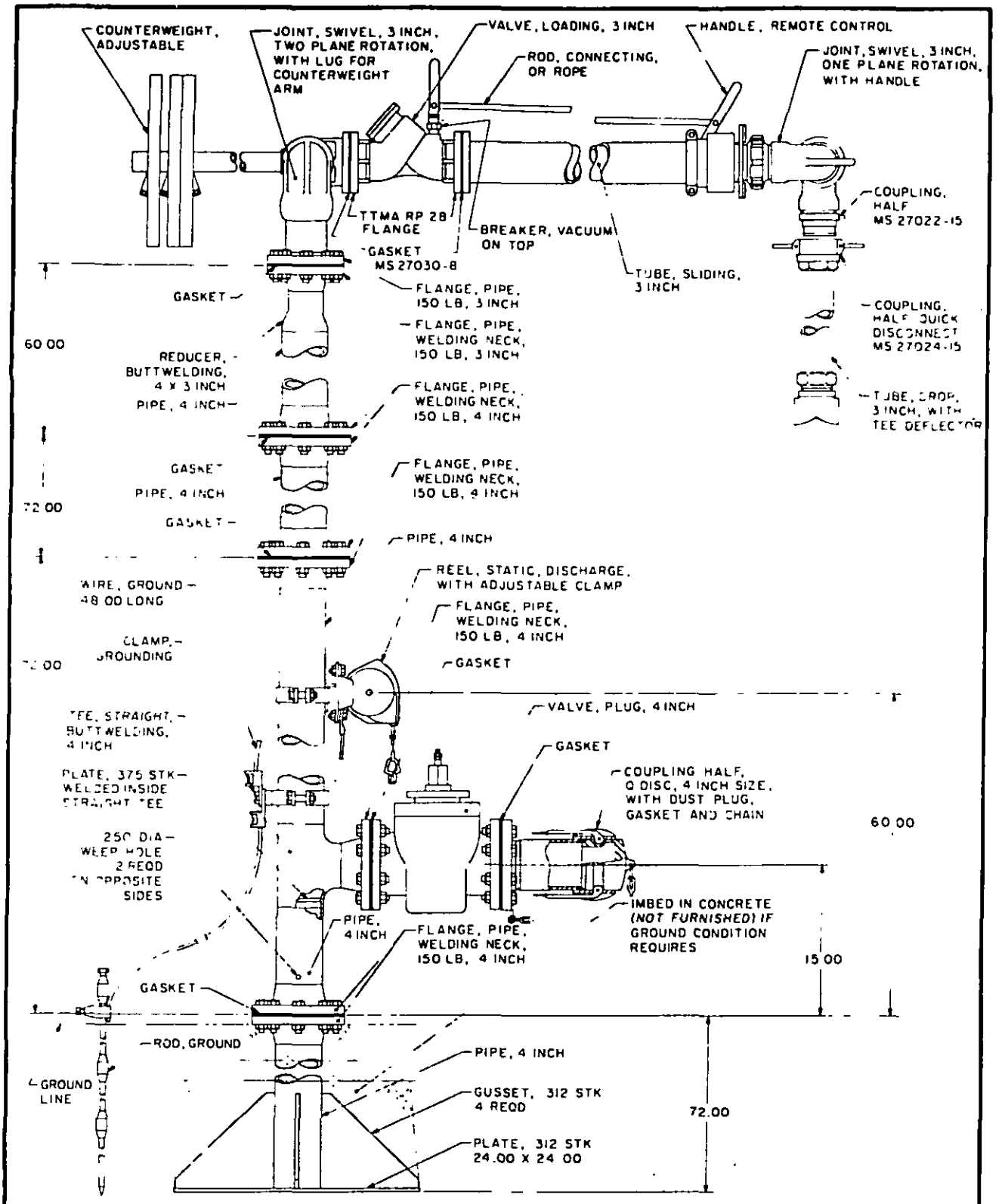


FIGURE 2. Loading standard, liquid transfer, bulk, sta, type II - sectional. X-2039B

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-L-3058G	2. DOCUMENT DATE (YYMMDD) 92/05/19

3. DOCUMENT TITLE
Loading Standards, Liquid Transfer, Bulk: Liquid Fuels, Stationary, 3-Inch

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. *PREPARING ACTIVITY*

a. NAME Violet Stewart	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3467 (2) AUTOVON 654-3467
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD FORM 1426 OCT 89 Previous editions are obsolete

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